

The Hydrogen Atom

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1 The Wave Equation

The wave equation,

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}, \quad (1)$$

can be derived from mechanical considerations including Newton's Second Law. All wave functions must be solutions to the wave equation. We are motivated to mathematically represent particles as waves, using their de Broglie wavelengths. By combining the wave equation with the Planck-Einstein relations;

$$E = \hbar\omega, \quad (2)$$

$$p = \hbar k, \quad (3)$$

we can express a particle as a wave of the form:

$$\psi(x, t) = Ae^{\frac{i}{\hbar}(px - Et)}. \quad (4)$$

We can find operators that return E and p as eigenvalues when applied to ψ , and combine them with the total energy relation for a bound particle $E = \frac{p^2}{2m} + V(x)$ to produce the time dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x)\psi(x, t). \quad (5)$$

By substituting the Hamiltonian operator and the energy eigenvalue we can get the time independent Schrödinger equation:

$$\hat{H}\psi = E\psi. \quad (6)$$

Because $|\psi|^2$ represents a probability, its integral over all space must be equal to 1. It can be shown through integration that if it is initially normalised, it remains normalised over time.

2 Boundary Conditions and the Infinite Square Well

To consider the potential function contribution to the Schrödinger equation, we first examine the most simple potential well possible, the infinite square well. This is a line with 0 potential within the bounds $0 \leq x \leq L$ and infinite potential outside. Assuming that ψ must be continuous and applying the boundary conditions, we find that only discrete energies and corresponding wave functions can exist in this system:

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2}, \quad (7)$$

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right). \quad (8)$$

In this case, n is the principal quantum number, which takes an integer value and represents the energy level of the system. Crucially, the discrete nature of the solution has arisen from the general wave solution and the boundary condition assumptions. Note that the ground state E_1 has a non-zero energy. The distribution approaches a uniform one as $n \rightarrow \infty$, indicating that the particle is equally likely to be found anywhere on the line.

3 The Harmonic Oscillator

The same analysis can now be applied to a harmonic oscillator. The harmonic oscillator is a useful model because any smooth potential near a stable equilibrium is approximately quadratic, so small oscillations are harmonic. Systems of coupled oscillators can then be decomposed by Fourier analysis into independent normal modes.

The potential function of a harmonic oscillator is:

$$V(x) = \frac{1}{2}m\omega^2 x^2. \quad (9)$$

By adopting a natural length scale $\xi = x/x_0$, a series solution can be found of the form:

$$\psi(x) = H(\xi)e^{-\frac{1}{2}\xi^2}, \quad (10)$$

with the power series coefficient relation:

$$a_{j+2} = \frac{2j - \varepsilon + 1}{(j+2)(j+1)} a_j, \quad (11)$$

where $\varepsilon = 2E/\hbar\omega$. The condition that ψ must be normalisable enforces a termination condition on this power series, and applying this produces the discrete energy levels that the system can occupy:

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega. \quad (12)$$

A consequence of the uncertainty principle is that the ground state, $E_0 = \frac{\hbar\omega}{2}$, is not zero. The same result can be achieved by considering ladder operators $\hat{a}_+$ and $\hat{a}_-$ to move up and down energy levels. This method is simpler than the series ODE method.

4 The Hydrogen Atom

The potential function of the hydrogen atom is known as the Coulomb potential:

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}. \quad (13)$$

Since the potential is spherically symmetric, we convert the Schrödinger equation into spherical coordinates. This involves first using the spherical Laplacian,

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi^2}, \quad (14)$$

which leads to the spherical time independent Schrödinger equation:

$$\hat{H}\psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(r)\psi. \quad (15)$$

The Laplacian can then be expanded and all terms can be separated into radial and angular parts. One side of the equation depends only on r and the other side depends only on θ and ϕ . Since they equal each other, both sides must be constant, allowing for the construction of a radial ODE and an angular ODE. The angular ODE can be further separated into an ODE in θ and one in ϕ . This separation is represented as $\psi(r, \theta, \phi) = R(r) \cdot Y(\theta, \phi)$ where $Y(\theta, \phi) = \Theta(\theta) \cdot \Phi(\phi)$.

The azimuthal angle equation is a second order homogeneous ODE, which has the solution $\Phi(\phi) = Ae^{im\phi}$, where m is a constant and A is a normalisation factor. The integer condition is forced because $\Phi(\phi) = \Phi(\phi+2\pi)$, as it represents an angle.

The polar angle equation simplifies with a substitution to the Legendre differential equation, which has a known solution in the form of an associated Legendre polynomial: $\Theta(\theta) = P_l^m(\cos \theta)$, where l is a constant. From forced termination conditions and the necessity of a non-zero polynomial, which arises from the wave function representation, the following constraints are introduced: $l \geq 0$, $m \leq |l|$ and m and l must be integers.

The combination of the two angular equations can be recognised as the spherical harmonic:

$$Y_l^m(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} \cdot P_l^m(\cos \theta) \cdot e^{im\phi}, \quad (16)$$

where the normalisation constant can be determined by integration. This solution does not depend on the Coulomb potential.

The radial equation is solved by considering the behaviour at extremes, as well as a power series component, keeping in mind that the solution must be normalisable as it represents a component of the wave function. The series can be recognised as proportional to an associated Laguerre polynomial, L_{n-l-1}^{2l+1} . The solution is:

$$R_{nl}(r) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]}} \cdot e^{-r/(na_0)} \cdot \left(\frac{2r}{na_0}\right)^l \cdot L_{n-l-1}^{2l+1}\left(\frac{2r}{na_0}\right). \quad (17)$$

The normalisation constant is again determined by integration. a_0 is the **Bohr radius**, which arises when considering the ground energy state and represents a characteristic length scale for the hydrogen atom. It is the most probable electron radius for the 1s state and has a value of $\approx 0.529 \text{ \AA}$. The actual energies for each energy level also arise from this solution and are found to be:

$$E_n = -\frac{m_e e^4}{32\pi^2 \epsilon_0^2 \hbar^2 n^2}, \quad (18)$$

which evaluates to $E_n \approx -13.6 \text{ eV}/n^2$. Crucially, it scales in an inverse proportion to n^2 . The termination requirement for the series solution also introduces constraints on n and l . The final constraints for all quantum numbers are shown below.

$$n, m, l \in \mathbb{Z} \quad n \geq l+1 \quad l \geq 0 \quad m \leq |l| \quad n \geq 1. \quad (19)$$

The numbers serve different functions. n determines the energy level, l determines the magnitude of the orbital angular momentum, and m determines the z-component of the angular momentum, which acts to orient it.

5 The Numerical Solver

The program uses a numerical solver for the radial equation, which depends on the Coulomb potential but has a complicated analytical solution. The interval between the boundaries $\psi(a) = \alpha$ and $\psi(b) = \beta$ is divided into N equal pieces, with a spacing of $h = (b-a)/N$. The three-point finite difference method is constructed from Taylor approximations at the grid points and used to convert the differential equation into a tridiagonal matrix equation. This is then solved in Python.

Using the wave vector, a discrete cumulative distribution function is constructed for the radial and angular distributions. This is then randomly sampled from to produce a cloud of possible locations for the electron, coloured by the relative probability of finding an electron at that point. The resulting visualisation can be adjusted to observe the dependence of the shape of the cloud on n , l , and m .